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Towards Intelligent Legal Advisors for Document Retrieval and Question-Answering in German Legal Documents

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***Abstract*—The legal system is one of the most important pillars of human society. While digitization is integrated into many areas of everyday life, the legal system is still very traditionally positioned. Recent advances in storing and processing large number of documents initiated the work of intelligent legal advisors. While first approaches in the English language suggest an enormous potential to generate knowledge from legal documents, there are no approaches in the German legal system. We present an intelligent legal advisor based on semantic document retrieval, to improve knowledge extraction from German legal documents. In addition, we set up a question-answering system. We implemented a BERT and a BM25-model for German document retrieval in legal documents. The approach is validated on a data set consisting of German question-answer pairs.**

***Index Terms*—NLP, knowledge extraction, question-answering, semantic search, document retrieval, German language**

I. Introduction

Legal documents are essential for our modern society and form a necessary foundation of legal information and democracy. In the course of digitization, the number of digitally stored legal documents increased considerably, and accordingly the need to process them more efficiently [1]. Recent advances in the areas of Natural Language Processing (NLP) and Knowledge Engineering have opened up, enabled new possibilities to assist law professionals and legal laypersons in processing and especially extracting information from legal documents in a more efficient way. This enables simplified and democratized access to legal information for the society [2]. These advances have led to first legal knowledge databases and other approaches processing legal documents. Unfortunately, the approaches available refer exclusively to documents from the English-speaking area, which differ fundamentally from German legal documents in their structure, linguistic conditions and the law itself [3]. This makes it necessary to develop a novel workflow for the processing of German legal documents, taking into account both linguistic peculiarities and structures as well as the need to use new technologies.

In this paper, we present our approach towards an intelligent legal advisor built on state-of-the-art technologies in the fields of: NLP, semantic search and knowledge engineering. Our workflow enables efficient document retrieval and question-answering (QA) within German legal documents, realized with German transformer-based language models BERT and ELECTRA, and a technology stack entirely based on open source software libraries [2], [4]–[6]. As a proof of concept, we provide results of the document retrieval in a corpus of German legal and German Wikipedia documents based on a pre-trained BERT language model and sparse retrieval methods.

II. Related Work

The *Caselaw Access Project (CAP)* aims to make all published U.S. court decisions from the Harvard Law School Library collection freely available in a uniform format. The documents are enriched with additional metadata and accessible using a free-text search [7]. *CourtListener* is a project similar to CAP launched at the University of California, Berkeley. The goal of the project is to provide free access to past and current court decisions of the United States legal system [8]. A project by Finland's Ministry of Justice is the *Finlex data base*, which makes legal documents publicly available in various data formats for machine processing and data analysis [9]. *Openlegaldata* is a platform from Germany providing free and public access to legal documents and aims to improve transparency in the legal services. For this purpose, the project collects legal data from various sources and provides them in machine-readable formats [10].

Another approach from the German area is Elena Leitner's published data set for *Fine-grained Named Entity Recognition in Legal Documents* which was created as part of the *Lynx project* and provides an annotated data set for the automated extraction of legal entities as well as first machine learning models [11], [12]. Moreover, there are already first approaches applying transformer-based models to the domain of law to solve various tasks such as the classification or generation of legal documents [13]–[16]. Unfortunately, they refer exclusively to English documents and cannot be easily adapted. Despite the already existing and promising solutions, there is still no system for semantic document retrieval and QA in German legal documents available [17]. Therefore, our research aims to close this gap and create a bridge between the field of German law and modern semantic search technologies.

III. The Intelligent Legal Advisor

The intelligent legal advisor consists of several components and is structured in different processing units, which are

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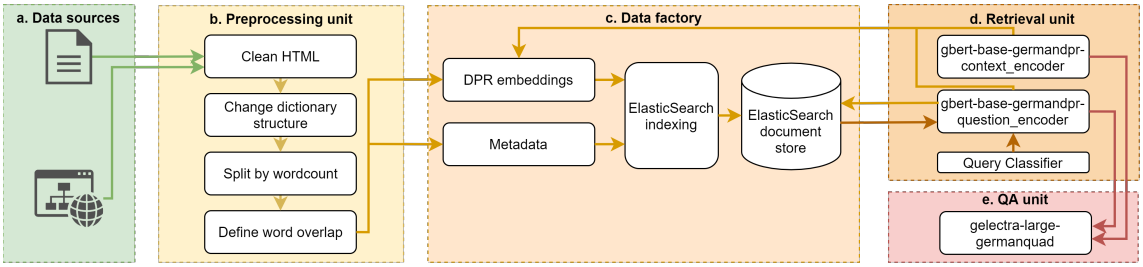


Fig. 1. Workflow of the legal advisor consists of various processing units that are processed sequentially

illustrated in Fig. 1. The basis are sufficient data sources (Fig. 1.a); our current database consists of approx. 200,000 German court rulings and decisions published by the open-data platform *Openlegaldata* [10]. The documents originate from a variety of courts at the city, state, and federal levels and were published between the years 1970 and 2021. Each document consists of the plain judgment text with additional metadata such as the assigned field of law or court.

The pre-processing unit (Fig. 1.b) is used to prepare incoming documents for further use. For this purpose, HTML elements are removed from the plain text and the formatting into the American Standard Code for Information Interchange (ASCII) format is performed. In addition, the documents are converted to a specific dictionary format, which consists of metadata and plain text. Furthermore, we divide the documents into passages with t=100 words each. This is necessary due to the fact that we use dense methods for the document retrieval process, which can process a maximum length of 256 tokens and have a faster processing speed for smaller passages [18]. To avoid losing important syntactic context when splitting documents into passages, we respect sentence boundaries and define an overlap of two words between each split.

The data factory (Fig. 1.c) consists of a database and a passage encoder model, which is responsible for indexing the pre-processed documents. Therefore, the text of each passage is stored as a passage vector in the database and is enriched with additional metadata. In our approach, we use the open source document-oriented database and search engine Elasticsearch. Elasticsearch enables fast and accurate document searching with various configuration options. In addition, support for dense retrieval methods is provided out of the box.

Finding relevant information in a corpus of documents is a complex task which can be solved using different approaches. The methods can be divided into sparse and dense methods, with the goal of representing words as vectors. One of the best known sparse methods is Term Frequency-Inverse Document Frequency (TF-IDF), which is a statistical measure for evaluating the relevance of individual terms in documents. For this purpose, TF-IDF forms sparse vectors from the given words, in which non-zero values within the vectors represent the number of unique words. An advantage of TF-IDF is that it consumes few resources and can be used without any training. Unfortunately, it does not take into account linguistic structures such as semantics or context of words. A further development of TF-IDF is BM25. An important advantage of BM25 is that the influence of the frequency of a term is weaker for high-frequency terms than for TF-IDF. In addition, the length of the documents is taken into account, which is one of the reasons why BM25 often provides better results than TF-IDF [19]. An alternative approach are dense methods to generate deep word vectors that map similarities between words to take semantic structures into account. For this purpose, neural networks are used to generate n-dimensional word vectors and represent them in a vector space. Well-known methods for generating vectors from words or documents are Word2Vec and Doc2Vec [20]. However, state-of-the-art transformer-based language models have the advantage of including multiple features in the generation of dense vector representations (e.g. attention-mechanism) [5].

In our approach the retrieval unit (Fig. 1.d) provides a lightweight and efficient way to search for documents in the database and extracts a set of k relevant documents based on an input query given by the user. For this matter we use dense passage retrieval (DPR) which was introduced by Facebook AI in 2020 as an alternative to sparse methods like BM25 [21]. DPR uses a dual encoding approach consisting of two BERT-based encoders. One encoder is responsible for encoding the question posed by the user, the other one for encoding the documents. The presented ranking of relevant documents to a question asked is calculated based on the dot product similarity between the query embedding and the passage embedding [22]. In doing so, DPR takes semantic similarities and structures as well as the context of words into account. Thus, the approach was able to outperform sparse methods in previous studies [21]. In our approach, both the legal documents and the questions asked by the user are encoded using pre-trained German BERT-models [23], [24]. Both models are trained on 9275 German question-answer pairs, while each pair consists of one positive passage and three negative passages taken from different German Wikipedia articles [25], [26]. Each input query made by the user is categorized by a query classifier as either a question or a keyword. If only keywords are detected, the QA unit (Fig. 1.e) is not activated and the k relevant passages retrieved will be passed directly to the user. Once a specific question is detected

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TABLE I
EXPERIMENT RESULTS WITH k=10 RETRIEVED PASSAGES

Retrieval Method	Recall	MAP	Percent
BM25, Legal data set	0.84	0.73	84%
DPR, Legal data set	0.25	0.10	25%
BM25, GermanQuAD	0.96	0.89	96%
DPR, GermanQuAD	0.82	0.60	82%

by the query classifier, the task of the QA-unit is to extract specific answers to the posed question from the k passages. For this purpose, a pre-trained, transformer-based ELECTRA model is used in our approach. ELECTRA models are an evolution of BERT, making the training process more efficient by introducing new training methods that outperform existing techniques such as the masked language modeling (MLM) [6]. The German ELECTRA model used was specifically trained for the extraction of answers in German text documents on a data set consisting of 11518 question-answer pairs [27].

IV. EXPERIMENTS

In this section, we conduct experiments to evaluate the performance of sparse and dense retrieval methods in the domain-specific environment of law.

A. Data set construction

To test the retriever performance in the legal domain, we created a hand-annotated data set of 200 question-answer pairs from German legal documents and structured it in the SQuAD format [28]. We emphasized that the questions asked refer to legal subjects such as paragraphs or legal references rather than general knowledge. By taking this approach to the annotation process, we ensured that the behavior of retrieval methods could be measured and compared specifically in the legal domain. To compare the domain-specific performance of the retrievers with more general documents, we further selected question-answer pairs from the GermanQuAD data set [25].

B. Experimental setup

All experiments are conducted using the BM25 and DPR retrieval methods on both the hand-annotated data set consisting of legal documents and the GermanQuAD data. For the DPR tests, we use the pre-trained BERT model mentioned in Section III. The created embeddings for each passage and input question are calculated with a dimension of d=768 and were stored in an Elasticsearch database. In total, we conducted two different experiments: In the first experiment, we evaluated both retrieval methods on the top-k=10 retrieved passages (Table I) and compared the retriever's results. Furthermore, we tested the relationship between the recall score as a function of the top-k retrieved passages.

C. Results

Table I shows the behavior of DPR and BM25 on our hand-annotated legal data set as well as on the GermanQuAD data. To evaluate our system, we calculated the recall score and the mean average precision (MAP). While the recall score describes the proportion of questions and correct passages in the k retrieved documents, the MAP considers higher rank of relevant documents within the top-k results. It can be shown that both BM25 and the pre-trained DPR model can retrieve relevant passages to the posed legal questions. Thereby, BM25 performs better than DPR on the legal documents in both recall and MAP. Compared to the GermanQuAD data, both BM25 and DPR have a recall score > 0.8 and show significantly better scores than on the legal data set.

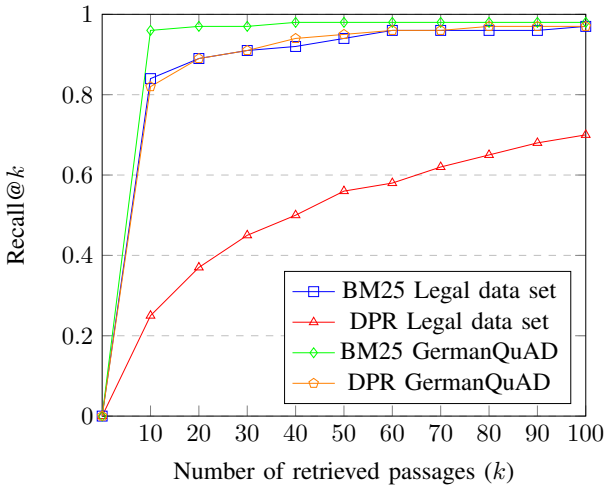


Fig. 2. Comparison of BM25 and DPR

Figure 2 illustrates the recall score of BM25 and DPR in relation to the top-k retrieved passages. We can state that both BM25 and DPR can improve recall in legal documents as the number of k passages increases, while on GermanQuAD the recall is already considerably high from the beginning.

V. CONCLUSION & FUTURE WORK

In this paper, we introduced our approach towards an intelligent legal advisor providing semantic document retrieval and QA in a corpus of German legal documents. We have shown that document retrieval and QA are highly relevant problems within the law that can be improved by our state-of-the-art technology approach, making the work of lawyers more efficient and lowering the barriers to society's access to legal information. With this initial work, we made first steps towards a holistic platform for information extraction and knowledge engineering in Germany's legal domain.

Furthermore, we described the workflow with its underlying technologies in detail and performed first experiments on document retrieval in the legal domain. Thereby we demonstrated that the pre-trained BERT model is not applicable to the domain of law out-of-the-box and performs worse than BM25 in both recall and MAP. Since the results of DPR were noticeably better on the GermanQuAD data set, we argue that a significant improvement can be achieved by fine-tuning the pre-trained model with a transfer learning approach. This leads to the future work of the project: (1) We want to increase

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our self-annotated data set of legal question-answer pairs and train the model taking into account the German legal structure and language dependencies. (2) In addition, we would like to implement and test the ELECTRA model for QA tasks in the legal sector and expand our data sources. To do so, we aim to access various official federal and state data sources and generate daily updated court rulings using official APIs and web scrapers. (3) To further facilitate the work of lawyers, we want to develop Named Entity Recognition methods for the automated extraction and summary of legal entities.

The presented data set and the corresponding code can be found at: github.com/Christoph911/AIKE2021_Appendix.

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